

Colony Benchmark Design (Proposed)

This supplementary section presents the *design specification* for colony cognitive security benchmarks introduced in Part 1, Section S05. These benchmarks are proposed for future implementation; the current CIF codebase validates individual-agent and small-group defense mechanisms (3–10 agents) as described in the main text. The configurations below define the target infrastructure for scaling CIF evaluation to colony-scale populations ($n > 10$).

Status: *The benchmark specifications in this section are proposed designs. The code snippets illustrate the intended API and are not yet implemented in the CIF repository. Colony-scale evaluation is an active area of future work (see sec:discussion).*

1. **Scalable agent populations** — $n \in \{10, 50, 100, 500, 1000\}$
2. **Configurable stigmergic substrates** — Shared memory, message queues, artifact stores
3. **Instrumented communication channels** — Full message logging with timestamps